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Open Information Extraction from 2007 to 2022 – A Survey

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Abstract

Open information extraction is an important NLP task that targets extracting structured information from unstructured text without limitations on the relation type or the domain of the text. This survey paper covers open information extraction technologies from 2007 to 2022 with a focus on new models not covered by previous surveys. We propose a new categorization method from the source of information perspective to accommodate the development of recent OIE technologies. In addition, we summarize three major approaches based on task settings as well as current popular datasets and model evaluation metrics. Given the comprehensive review, several future directions are shown from datasets, source of information, output form, method, and evaluation metric aspects.

1 Introduction

Open Information Extraction (OIE) aims to extract structured information from unstructured text (Niklaus et al., 2018), typically on the form of a relation between two entities as a triple ($entity_1, relation, entity_2$). OIE remains to be a popular topic among AI researchers because it is an important upstream task in machine reading and understanding with its output contributing to numerous natural language processing (NLP) tasks such as question answering, search engine, and knowledge graph completion (Han et al., 2020). There are also outputs other than relations (Zouaq et al., 2017), but the amount of work is relatively small. Thus this survey focus on *open relation extraction*.

Work on OIE dates back to 2007. The first generation of OIE models exemplifies TEXTRUNNER (Banko et al., 2007), WOE (Wu and Weld, 2010)

and REVERB (Fader et al., 2011), which use shallow linguistic features such as part-of-speech (POS) tags and noun phrase (NP) chunk features. The second generation of OIE models, represented by OLLIE (Schmitz et al., 2012), ClausIE (Del Corro and Gemulla, 2013), SRL-IE (Christensen et al., 2010) and OPENIE4 (Mausam, 2016) makes use of deep linguistic features obtained in addition to shallow syntactic features. The third generation of OIE models, which we will thoroughly discuss, is deeply influenced by the emergence of neural models such as Transformers (Vaswani et al., 2017), especially BERT (Devlin et al., 2019), and makes heavy use of pre-training models to produce features.

Previous surveys on OIE systems (Vo and Bagheri, 2018; Zouaq et al., 2017; Glauber and Claro, 2018; Niklaus et al., 2018) cover OIE models from 2007 to 2018 and commonly view models from the method perspective. An early OIE overview Gamallo (2014) divides OIE systems into four groups in two steps as shown in figure 1: first, systems are divided into data-based¹ and rule-based categories based on whether hand-crafted rules are needed; then, each category is split in half based on whether the system uses shallow syntactic analysis or dependency parsing. For a further reference to previous important OIE models and surveys, readers can look at the timeline in figure 2, where the introduction of the new source of information in OIE and existing essential reviews are shown. However, no survey is available after 2018.

The rise of pre-training techniques makes the categorization of Gamallo (2014) difficult for covering recent work. In this survey, we propose a new categorizing method for OIE systems from the source of information perspective in Section 3. A comparison of our grouping method with Gamallo (2014) is pictured in figure 1. By analyzing the

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¹We use "Data-based" instead of "Training-data" in original paper (Gamallo, 2014) for better understanding.

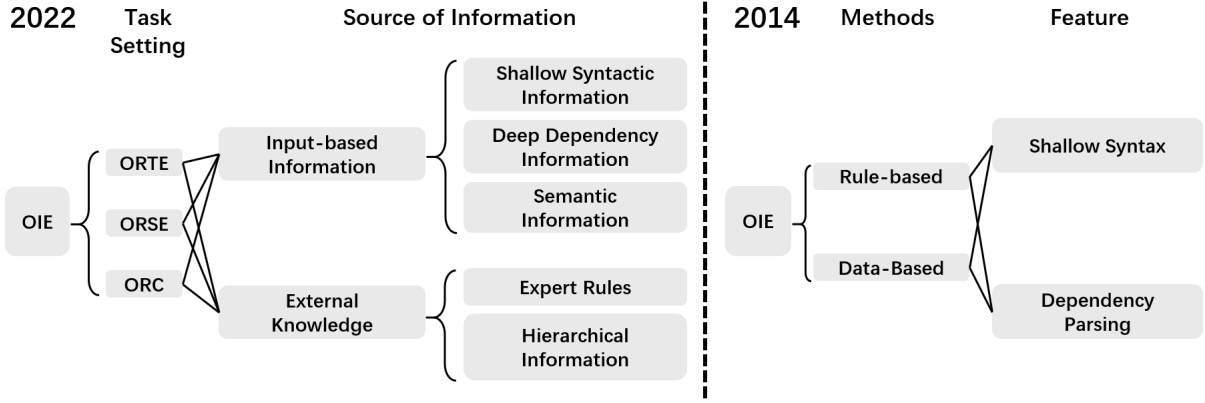


Figure 1: OIE system categories in 2014 and 2022. **ORTE**, **ORSE** and **ORE** are short for Open Relational Triple Extraction, Open Relation Span Extraction, Open Relation Clustering respectively, which will be introduced in Section 4 in detail. Braces include subcategories of a class, and string connections represent information sources or features that can be used by a certain type of model.

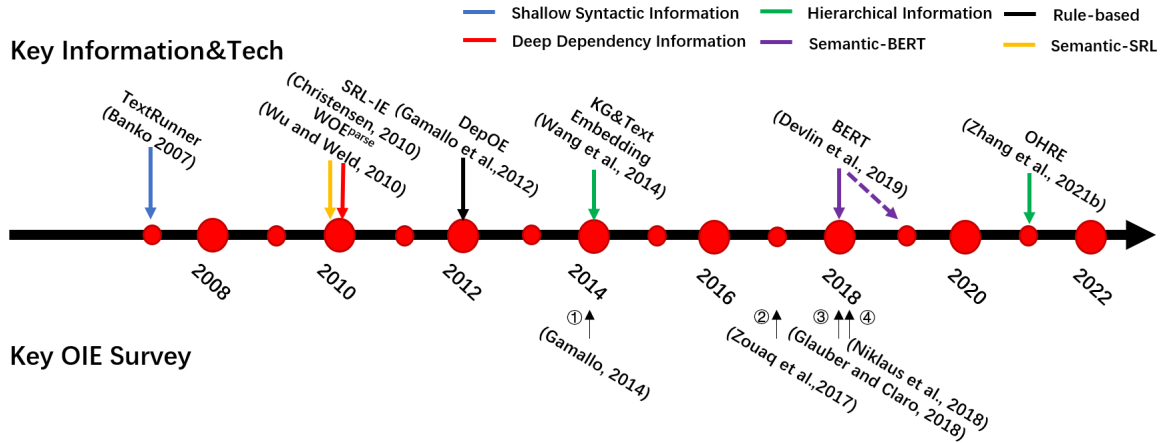


Figure 2: Timeline of the information sources and key technologies and related OIE surveys. Above the timeline axis is the time when information sources and key technologies were first used in OIE, while Below is the important survey of OIE. The dash arrow shows BERT-based methods are widely used since 2019.

usage of information, we show the development of OIE systems as the techniques advance. Moreover, we introduce new OIE models that are not included in the old OIE surveys in Section 4. To give readers a comprehensive view, we group recent OIE models based on their task settings because their sources of information do not differ much. We summarize three major approaches—Open Relational Triple Extraction, Open Relation Span Extraction, and Open Relation Clustering—to the OIE problem, with a few models following into the other categories. In addition, this paper summarizes current popular datasets and model evaluation metrics in Section 2 and Section 5, respectively, as a reference. We suggest several future directions from datasets, source of information, output form, method, and evaluation metric perspectives in Section 6.

2 Datasets

Previous surveys (Niklaus et al., 2018; Ali et al., 2019) conclude with models of the first two generations and the datasets they used. We exclude those small-scale datasets and some seldom-used datasets. In table 1, we list some popular and promising OIE datasets grouped by their creating methods.

The first group is converted from other crowdsourced Question Answering (QA) datasets. OIE2016 (Stanovsky and Dagan, 2016) is one of the most popular OIE benchmarks, which leverages QA-SRL (He et al., 2015) annotations. AW-OIE (Stanovsky et al., 2018) extends the OIE2016 training set with extractions from QAMR dataset (Michael et al., 2017). The OIE2016 and AW-OIE datasets are the first datasets used for supervised OIE. However, because of its coarse-grained gener-

Dataset	#Sent.	#Tuple	Domain
OIE2016	3200	10,359	Wiki, Newswire
Re-OIE2016	3200	NR	Wiki, Newswire
CaRB	3200	NR	Wiki, Newswire
AW-OIE	3300	17,165	Wiki, Wikinews
LSOIE-wiki	24,296	56,662	Wiki, Wikinews
LSOIE-sci	47,998	97,550	Science
WiRe57	57	343	Wiki, Newswire
SAOKE ^{zh}	48,248	NR	Baidu Baike
BenchIE ^{en}	300	136,357	Wiki, Newswire
BenchIE ^{de}	300	82,260	Wiki, Newswire
BenchIE ^{zh}	300	5,318	Wiki, Newswire
NYT-FB	1.8M	39,000	NYT, Freebase
FewRel	NR	70,000	Wiki, Wikidata
T-REx SPO	763,000	NR	Wiki, Wikidata
T-REx DS	12M	NR	Wiki, Wikidata
COER ^{zh}	NR	1M	Baidu Baike, Chinese news

Table 1: Statistics of popular OIE datasets. ("NR" stands for "Not Reported". Multilingual and non-English datasets are indicated with superscripts.)

ation method, OIE2016 has some problematic annotations and extractions. On the basis of OIE2016, Re-OIE2016 (Zhan and Zhao, 2020) and CaRB (Bhardwaj et al., 2019) re-annotate part of the dataset. LSOIE (Solawetz and Larson, 2021) is created by converting QA-SRL 2.0 dataset (FitzGerald et al., 2018) to a large-scale OIE dataset, which claims 20 times larger than the next largest human-annotated OIE dataset.

The second group is directly crowdsourced, including WiRe57 (L  chelle et al., 2019), SAOKE dataset (Sun et al., 2018), and BenchIE dataset (Gashteovski et al., 2021). WiRe57 is created based on a small corpus containing 57 sentences from 5 documents by two annotators following a pipeline. SAOKE dataset is generated from Baidu Baike, a free online Chinese encyclopedia, like Wikipedia, containing a single/multi-span relation and binary/polyadic arguments in a tuple. It is built in a predefined format, which assures its completeness, accurateness, atomicity, and compactness.

The third group is established by aligning triples in the knowledge base (KB) with text in the corpus. Several works (Mintz et al., 2009; Yao et al., 2011) have aligned the New York Times corpus (Sandhaus, 2008) with Freebase (Bollacker et al., 2008) triples, resulting in several variations of the same dataset, NYT-FB. FewRel (Han et al.,

2018) is created by aligning relations of given entity pairs in Wikipedia sentences with distant supervision, and then filtered by human annotators. ElSahar et al. (2018) propose a pipeline to align Wikipedia corpus with Wikidata (Vrande    , 2012) and generate T-REx. By filtering triples and selecting sentences, Hu et al. (2020) create T-REx SPO and T-REx DS.

Different from all other datasets, COER (Jia et al., 2018) is a Chinese dataset automatically created by an unsupervised open extractor from diverse and heterogeneous web text, including encyclopedia and news. Among those datasets, the third group is mostly used in open relation clustering task setting, illustrated in Section 4.3. Whereas others are usually used in open relational triple extraction (Section 4.1) and open relation span extraction task settings (Section 4.2).

3 Source of information

We categorize information sources into two groups: *input-based information* and *external information*. Input-based information is defined as information explicitly or implicitly contained in the input unstructured text. Implicit input-based information, typically semantic information, is represented as vectors by the embedding layer of OIE whereas explicit information such as shallow syntactic information needs to be extracted with the help of a parser. External information includes information that is not contained in the input text. Usually, external information is used in OIE systems as a supplement to input-based information to improve the model performance. Examples of external information are predefined rules and knowledge bases.

3.1 Input-based Information

Shallow syntactic information such as part of speech (POS) tags and noun-phrase (NP) chunks abstract input sentences into patterns. It is pervasively used in the early work of OIE as an essential model feature (Banko et al., 2007; Wu and Weld, 2010; Fader et al., 2011). In rule-based models, those patterns directly determine whether the input text contains certain relations or not (Xavier et al., 2013; A and A, 2013). Shallow syntactic information is reliable because there is a clear relationship between the relation type and the syntactic information in English (Banko et al., 2007). However, merely using shallow syntactic information can not discover all relation types. Subsequent work uses

shallow syntactic information as part of the input and incorporates additional features to enhance the model performance (Stanovsky et al., 2018).

Deep dependency information shows the dependency between words in a sentence, which can be used directly to find relations (Vo and Bagheri, 2018). But because dependency analysis is more complex and time-consuming than shallow syntactic analysis, such information source was not popular in early OIE studies. It was the second generation of OIE models that brought dependency parsing to great attention. Right now, dependency information is still used as part of the model input, though with less popularity and sometimes not directly. Elsayar et al. (2017) make use of the dependency path to give higher weight to words between two named entities, in which way the model only uses dependency information as a supplement and relies more on the semantic meaning to extract information.

Semantic information captures not only linguistic structures of sentences but literal meanings of phrases, which can express more diverse and fitting relations compared to syntactic patterns. However, semantic information can also be too specific and hence lead to the canonicalizing problem (Galárraga et al., 2014; Vashishth et al., 2018; Wu et al., 2018). The second generation of OIE models has tried to use semantic information via semantic role labeling, for example EXAMPLAR (Mesquita et al., 2013), or via dependency parsing, for instance OLLIE (Schmitz et al., 2012). There were also attempts to use WordNet output to comprise semantic information (Liu and Yang, 2012). The third generation of OIE models typically use the word and sentence representations obtained from pre-trained language models (Kolluru et al., 2020b; Hwang and Lee, 2020; Xinwei and Hui, 2020). These representations contain both syntactic and semantic information (Jawahar et al., 2019). Meanwhile, some OIE models use word embeddings from word embedders such as GloVe, ELMo, and Word2Vec to capture semantic information (Ni et al., 2021).

3.2 External Knowledge

Expert rules are knowledge imported in the form of heuristic rules. It is easy for rule-based OIE systems to incorporate domain knowledge as well as to trace and fix errors (Chiticariu et al., 2013). Heuristic rules can be employed to avoid incoherent

extractions (Fader et al., 2011). For example, verb words between two entities are likely to be the relation. Thus, to alleviate incoherence, a rule can be defined: *If there are multiple possible matches for a single verb, the shortest possible match is chosen.* Based on patterns generated from POS-tagging, dependency parse, and other syntactic analyses, different rules can be created.

Hierarchical information that implicitly exists in languages, which can be explicitly exhibited by knowledge bases, benefits knowledge representation learning (Wang et al., 2014; Lin et al., 2015; Hu et al., 2015; Xie et al., 2016). In addition, KBs contain fine-grained factual knowledge that provides background information and hierarchical structures needed for relation extraction. Compared to traditional clustering, KB can provide hierarchical information that helps represent and cluster relations in a more organized way (Zhang et al., 2021) and hierarchical factual knowledge for data augmentation (Fangchao et al., 2021).

4 Task Settings

Banko et al. (2007) first define open information extraction as an unsupervised task to automatically extract $(entity_1, relation, entity_2)$ triples from a large amount of unstructured text on the web, where *relation* is composed of selected words in input sentences. Later, researchers have tried to solve OIE tasks under different conditions, leading to different task settings with various formats of input and output. In this section, we organize our subsections by task settings to introduce recent important OIE models since 2018. Three task settings are identified, and the two revised settings are named with corresponding approaches for better understanding. For an overall review, a table comparing significant new OIE models with classic models is available in the Appendix A.

4.1 Open Relational Triple Extraction

Task : $text \rightarrow relational\ triple$

In seminal work (Banko et al., 2007), the task setting was to extract entities relation triples $(entity_1, relation, entity_2)$ from unstructured text. In this setting, some approaches directly extract triples from text, while others take a two-step procedure, first find predicates, and then extract arguments. See Figure 3 for an example.

Direct Extraction. A typical method solves this

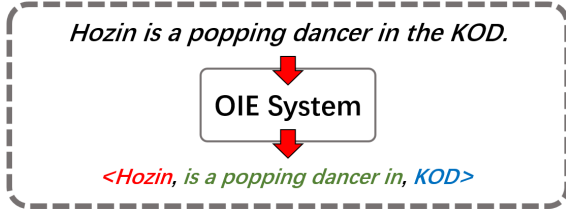


Figure 3: An example of open relational triple extraction.

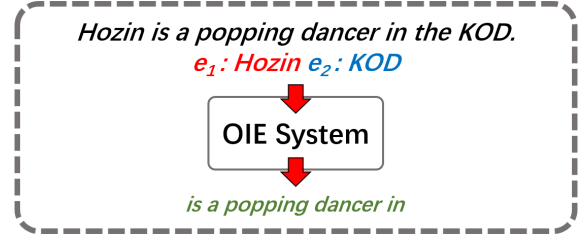


Figure 4: An example of open relation span extraction.

task from a *labeling* perspective. SenseOIE (Roy et al., 2019) improves upon RnnOIE (Stanovsky et al., 2018) by using the extractions of multiple OpenIE systems as features together with a manually annotated dataset. DetIE (Vasilkovsky et al., 2022) is a Transformer-based encoder-only model that extracts a large predefined number of triples from the free text by labeling each token with an artificial category. They unfreeze several top layers of the BERT encoder to extract semantic features as well as inter-token dependencies. OpenIE6 (Kolluru et al., 2020a) adopts a novel Iterative Grid Labeling (IGL) architecture, with which OpenIE is modeled as a 2-D grid labeling problem. Each extraction corresponds to one row in the grid. Iterative assignments of labels assist the model to capture dependencies among extractions without re-encoding.

Another popular paradigm to extract relational triples is *sequence generation*, which can produce auxiliary words or change the word order with the encoder-decoder framework. NeuralOIE (Cui et al., 2018) formulates the relation extraction task into a sequence-to-sequence task. To enlarge the vocabulary and reduce the proportion of generated unknown tokens, NeuralOIE uses the coping mechanism to copy words from input to output. IMoJIE (Kolluru et al., 2020b) is another generative approach, which iteratively generates extraction and appends it to the end of the input sequence until *EndOfExtractions* token is yielded. Thus each extraction is conditioned on all its preceding extractions. Aiming to extract triples from Chinese text, Sun et al. (2018) propose the Logician model, which relies on global coverage attention and gated-dependency attention, a language-specific heuristic for Chinese.

Apart from these two mainstream perspectives, Zhan and Zhao (2020) propose SpanOIE, a *span-based* model that has two sub-modules, the predicate module, and the argument module, both of which represent span features by calculating hid-

den states of the start token and the end token. The predicate module predicts whether the span is a predicate or not, and the argument module obtains the argument labels.

Two-step Extraction. Some researchers regard the relational triple as two non-empty arguments connected by a single predicate, thus formulating the triple extraction task into a two-step process. They first identify predicates as relational words and then use sequence labeling to obtain arguments. Stanovsky et al. (2018) introduce a novel BIO tagging scheme, which arranges arguments regardless of their order in the original sentence. They propose RnnOIE, a Bi-LSTM transducer followed by a Softmax classifier to extract multiple overlapping triples with given predicates and sentences. Multi²OIE (Ro et al., 2020) is a slightly different version of RnnOIE. Its first step is to label all predicates upon BERT embedded hidden states instead of locating predicates with syntactic features. The second step is to extract the arguments associated with each identified predicate by using a multi-head attention mechanism.

4.2 Open Relation Span Extraction

Task : $entities + text \rightarrow relation\ span$

Different from the previous setting, some methods find relational span according to the given argument tuple and sentence. See Figure 4 for an example. QuORE (Yang et al., 2022) is a framework to extract single/multi-span relations and detect non-existent relationships, given an argument tuple and its context. The model uses a manually defined template to map the argument tuple into a query. It concatenates and encodes the query together with the context to generate sequence embedding, with which this framework dynamically determines a sub-module (Single-span Extraction or Query-based Sequence Labeling) to label the potential relation(s) in the context.

Jia et al. (2022) propose a hybrid neural network model (HNN4ORT) for open relation tagging. The

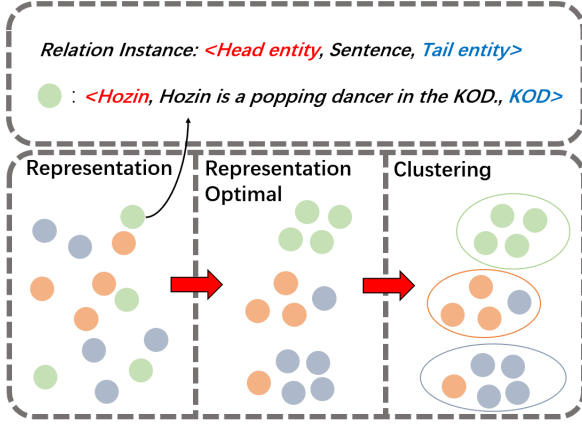


Figure 5: An overview of open relation clustering. Each node denotes a relation instance while different colors denote different relation classes.

model employs the Ordered Neurons LSTM (Shen et al., 2019) to encode potential syntactic information for capturing associations among arguments and relations. It also adopts a novel Dual Aware Mechanism, integrating Local-aware Attention and Global-aware Convolution.

4.3 Open Relation Clustering

Task : entities + text $\rightarrow$ clustering without explicit relation span or label

Open relation clustering (ORC), also widely called open relation extraction, clusters relation instances (h, t, s) , where h and t denote head entity and tail entity respectively, and s denotes the sentence corresponding to two entities. Different from the initial OIE task setting, ORC does not extract entities from text but uses the whole text to represent the relation between two entities. ORC models cluster similar relations, which is a step forward in labeling specific relations to each relation instance. See Figure 5 for an overview of this task setting.

Representation by Defined Features. Early clustering methods represent relation instances in feature space with the help of explicit features from various types of information. Ru et al. (2017) compares the contributions of different sequential patterns, syntactic information and the combination of the above information to the representation of relation instances, which are clustered by a hierarchical clustering algorithm (Zhao et al., 2005). The result shows that sequential patterns and syntactic information are both beneficial to relation representation.

Lechevrel et al. (2017) select core dependency phrases to capture the semantics of the relations

between entities. The design rules are based on the length of dependency phrase in the dependency path, which sometimes contains more than one dependency phrase that uses all terms and brings in irrelevant information. Each relation instance is clustered on the basis of the semantics of core dependency phrases. Finally, clusters are named by the core dependency phrase most similar to the center vector of the cluster.

Instead of directly cutting less irrelevant information, Elsahar et al. (2017) propose a more resilient approach based on the shortest dependency path. The model generates representations of relation instances by assigning a higher weight to word embedding of terms in the dependency path and then reduces feature dimensions by PCA (Shen, 2009). Although the model ignores noisy terms in the dependency path, re-weighting is a forward-looking idea resembling the subsequent attention mechanism.

With the development of pre-trained language models, contextualized semantics can be better represented. Before the clustering step, recent clustering-based approaches tend to optimize relation representations with different supervision signals instead of manually extracting features based on different rules.

Semantic Representation by PLM. Unlike the above OIE systems that follow Banko et al. (2007) to use unsupervised learning methods, RSN (Wu et al., 2019) exploits existing labeled data and relational facts in knowledge bases during training. To narrow the gap between pre-defined relations and novel relations, RSN learns a relational similarity matrix that can transfer relation knowledge from supervised data to unsupervised data. Finally, because high dimensional non-linear representations of relations are not suitable for calculating centroids by sum-average arithmetic, relations are clustered by Hierarchical Agglomerative Clustering and Louvain (Blondel et al., 2008).

Without taking advantage of labeled datasets, SelfORE (Hu et al., 2020) proposes a self-supervised learning method for learning better feature representations for clustering. SelfORE is composed of three sections: (1) encode relation instances by leveraging BERT (Devlin et al., 2019) to obtain relation representations; (2) apply adaptive clustering based on updated relation representations from (1) to assign each instance to a cluster with high confidence. In this way, pseudo labels

are generated. (3) pseudo labels from (2) are used as supervision signals to train the relation classifier and update the encoder in (1). Repeat (2).

As mentioned above high-dimensional vectors need to be clustered in a more complex way, Zhao et al. (2021) argue that such high-dimensional vectors contain too much irrelevant information for relation clustering, such as complex linguistic information. They propose a relation-oriented model based on SelfORE with a similar unsupervised training part and a modified supervised part. RoCORE leverages label data to learn low-dimensional relation-oriented vectors that can better reflect the category relationship through the distance between relation vectors. In this way, relation clustering performs very well and does not need to use complex clustering algorithms.

Hierarchical Information. Apart from labeled data, knowledge bases also benefit OIE by generating positive and negative instances. Datasets generated from distant supervision bring in spurious correlations (Roth et al., 2013; Jiang et al., 2018; Chen et al., 2021). Fangchao et al. (2021) conduct interventions to entities and context separately to avoid the spurious correlations of them to relation types. The key idea is based on blocking backdoor paths from a causal view (Pearl, 2000). The intervened context is generated by a generative PLM, while entities are intervened by placing them with three-level hierarchical entities in KB. Model parameters are optimized by those intervened instances via contrastive learning. The learned model encodes each instance into its representations, before using clustering algorithms.

Whereas the previously mentioned OIE system mainly focuses on optimizing representations of relation instances for better clustering, it is still unclear what relation each cluster represents after clustering. OHRE (Zhang et al., 2021) first proposes a top-down hierarchy expansion algorithm to cluster and label relation instances under the hierarchical structure of KB. Zhang et al. argue that the distance between entities in the hierarchy should reflect their semantic similarity. They design a dynamic hierarchical triplet objective to learn relation representations and unlabeled relation clusters are labeled with the hierarchical similarity of KB. In this way, clusters of existing relations are labeled clearly, and novel relations can be labeled as children relations of existing relation labels.

Task Setting	Evaluation Metrics
ORTE	Recall, AUC, F1
ORSE	F1
ORC	ARI, B^3 , V-measure

Table 2: The main evaluation metrics of each task setting. Full table contained model, method, source of information, dataset and result are shown in Appendix.

4.4 Other Settings

Translation. Wang et al. (2021a) cast information extraction tasks into a text-to-triple translation problem. They introduce DEEPEX, a framework that translates NP-chunked sentences to relational triples in a zero-shot setting. This translation process consists of two steps: generating a set of candidate triples and ranking them.

Multilingual. MILIE (Kotnis et al., 2022) is an integrated model of a rule-based system and a neural system, which extracts triple slots iteratively from simple to complex, conditioning on preceding extractions. The iterative nature guarantees the model to perform well in a multilingual setting. Multi²OIE (Ro et al., 2020) also has a multilingual version based on multilingual-BERT, which makes it able to deal with various languages. Differently, LOREM (Harting et al., 2020) trains two types of models, language-individual models, and language-consistent models and incorporates multilingual, aligned word embeddings to enhance model performance.

Noun Phrase ORE. ZORE (Qiu and Zhang, 2014) has explored Chinese noun phrase (NP) in a classical setting and achieved great performance. NPORE extracts relation triplets from Chinese NP (Wang et al., 2021b) instead of sentences because Chinese NPs usually omit clear predicates, making NP-based ORE in Chinese more difficult.

5 Evaluation

Model evaluation metrics used in OIE models differ depending on task settings. Models in open relational triple extraction setting (Section 4.1) and open relation span extraction setting (Section 4.2) mostly use precision, recall, F1 score, and area under PR curve (AUC), which might be calculated by different scoring functions. In open relation clustering setting, B^3 (Bagga and Baldwin, 1998), V-measure (Rosenberg and Hirschberg, 2007) and ARI (Hubert and Arabie, 1985) are used to evaluate the performance of clustering models.

Token-level Scorers. To allow some flexibility (e.g., omissions of prepositions or auxiliaries), if automated extraction of the model and the gold triple agree on the grammatical head of all of their elements (predicate and arguments), OIE2016 (Stanovsky and Dagan, 2016) takes it as matched. L  chelle et al. (2019) penalize the verbosity of automated extractions as well as the omission of parts of a gold triple by computing precision and recall at token-level in WiRe57. Their precision is the proportion of extracted words that are found in the gold triple, while recall is the proportion of reference words found in extractions. To improve token-level scorers, CaRB (Bhardwaj et al., 2019) computes precision and recall pairwise by creating an all-pair matching table, with each column as extracted triple and each row as gold triple.

Fact-level Scorers. SAOKE (Sun et al., 2018) measures to what extent gold triples and extracted triples imply the same facts and then calculates precision and recall. BenchIE (Gashteovski et al., 2021) introduces *fact synset*: a set of all possible extractions (i.e., different surface forms) for a given fact type (e.g., VP-mediated facts) that are instances of the same fact. It takes the informational equivalence of extractions into account by exactly matching extracted triples with the gold fact synsets.

6 Conclusion and Future Direction

From a chronological perspective, we see a trend of incorporating more diverse sources of information as well as a growing diversion in the approaches. These changes could be explained by the following points. The most important factor is the development of techniques in the deep learning field, especially the pre-training models which facilitate obtaining rich information from the input text. The computing power has also accelerated during the recent two decades, which supports the usage of more sources of information and more complex models.

OIE datasets have become larger and more suitable for this specific problem. But compared to the amount of information on the web, the size of existing datasets is still too small to be “open”. The domain of datasets is limited to Wiki, Newswire, NYT, and Freebase, although a few datasets contain corpus from different sources and in other languages. OIE datasets should be larger and include more languages in the future. In addition, unifica-

tion of test set segmentation is imminent to reduce the difficulty in cross-model comparisons.

From the **source of information** perspective, recent neural models rarely incorporate dependency information that is extracted by the parser, which is used in before-neural models as explicit constraints of sentence structure. However, it has been seen that dependency syntax benefits neural relation extraction (Li et al., 2017; Zhang et al., 2018; Li et al., 2021). So it is worth investigating dependency information.

Overly-specific relation output has always been a problem. Current OIE models still face challenges to extract succinct but meaningful relations. For sequence labeling and sequence generation models, current evaluation metrics might be a catalyst to this problem. These evaluation methods calculate the similarity between the extracted triples and gold triples from the token aspect instead of the semantic meaning aspect, hence resulting in verbosity and incompleteness of the model output. Therefore, as a potential research direction, inventing a more sophisticated semantic-level evaluation metric is likely to prompt the occurrence of better OIE models. On the other hand, overly-specific issue in ORC has a different cause. Relations in ORC are presented by the whole sentence and entities, but one sentence can contain more than one relations, which can introduce noise to the extracted relation. Noisy relation instances not only affect clustering performance but also bring inconvenience to downstream tasks. Therefore, a future work direction is to purify input text with respect to different relations before clustering.

Most clustering OIE models use K-means as the **clustering method**, which is problematic because prior knowledge about the number of novel relation classes is not available in real-world scenarios. Therefore, it is promising to focus on methods that can self-determine the number of clusters. Furthermore, alleviating the gap between relation representations and clustering methods also counts.

There still lacks a common **standard form for OIE output**, causing trouble for canonicalizing the output as well as agreeing on the golden standard. This also makes model comparison difficult. It can be useful if a set of standards can be formulated based on the requirements of different downstream tasks in the future.

Future research could also work towards a hybrid direction of supervised and unsupervised models

to further improve model performance. Trained using information from labeled data, unlabeled data, KBs, free text, and prior knowledge, the future OIE system should be able to efficiently and correctly extract relations of known types as well as extract and accurately label unseen relations given all available resources.

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A Appendix

Model	Method	Source of Information	Task Setting	Dataset	Evaluation (Result)
TEXTRUNNER (Banko et al., 2007)	Dependency Parser, NP Chunker, CRF, Naive Bayes Classifier	syntactic, dependency	4.1	400 Web	Average Error Rate (12%)
WOE (Wu and Weld, 2010)	TEXTRUNNER, Self-supervised Learning	syntactic, dependency	4.1	300 news 300 Wikipedia 300 Web	Precision-Recall Curve
REVERB (Fader et al., 2011)	Syntactic Constraints, Lexical Constraints, CRF	syntactic, dependency	4.1	500 Web	Precision-Recall Curve, AUC (1.3*WOE ^{parse} , 2*TEXTRUNNER)
OLLIE (Schmitz et al., 2012)	REVERB, Bootstrap, Open Pattern Learning	syntactic, dependency	4.1	300 news (from WOE) 300 Wikipedia (from WOE) 300 biology	Precision-Yield Curve, AUC (1.9*WOE ^{parse} , 2.7*REVERB)
OPENIE4 (Mausam, 2016)	SRLIE (Christensen et al., 2011), RELNOUN (Pal et al., 2016)	syntactic, dependency	4.1	Not Reported	Precision-Yield Curve, AUC (1.32*OLLIE, 4*REVERB)
ClausIE (Del Corro and Gemulla, 2013)	Dependency Parser, Clause-based Model	syntactic, dependency	4.1	500 Web (from REVERB) 200 Wikipedia 200 news	Precision-Yield Curve, # of correct extractions / # of extractions
RnnOIE (Stanovsky et al., 2018)	Bi-LSTM, Softmax	word emb, POS emb	4.1	OIE2016 WEB NYT PENN	AUC (48), F1 (62) AUC (47), F1 (67) AUC (25), F1 (35) AUC (26), F1 (44)
NeuralOIE (Cui et al., 2018)	LSTM, Copy Attention	word emb	4.1	OIE2016	AUC (27)
IMoJIE (Kolluru et al., 2020b)	BERT, LSTM, CopyAttention	word emb	4.1	CaRB	AUC (33.3), F1 (53.5)
SpanOIE (Zhan and Zhao, 2020)	Bi-LSTM, Span-consistent Greedy Search	word emb, POS emb, dependency relation emb	4.1	OIE2016 Re-OIE2016	AUC (48.9), F1 (68.65) AUC (65.9), F1 (78.50)
Multi ² OIE (Ro et al., 2020)	BERT, Multihead Attention	word emb, position emb, avg vector of predicates	4.1	Re-OIE2016 CaRB	AUC (74.6), F1 (83.9) AUC (32.6), F1 (52.3)
OpenIE6 (Kolluru et al., 2020a)	Iterative Grid Labeling, BERT, Self-attention	word emb, dependency feature	4.1	CaRB	AUC (33.7), F1 (52.7)
HNN4ORT (Jia et al., 2022)	ON-LSTM, CNN, Attention	word emb, POS emb, argument emb, local/global features	4.2	Wikipedia NYT Reverb	F1 (79.8) F1 (74.5) F1 (81.7)
UORE (Elsahar et al., 2017)	Re-weight Word Emb, TF-IDF, PCA, HAC	word emb, dependency	4.3	NYT-FB	F1 (41.6)
RSN (Wu et al., 2019)	Relational Siamese Network, CNN, HAC, Louvain	word emb	4.3	FewRel	B ³ : P (48.9) R (77.5) F1 (59.9)
SelfORE (Hu et al., 2020)	Bootstrapping Self-supervision, BERT, K-means, Adaptive Clustering	word emb	4.3	NYT+FB	ARI (40.3), B ³ : P (49.1) R (47.3) F1 (51.1), V: F1 (46.6) Hom (45.7) Comp (47.6)
				T-REx SPO	ARI (33.7), B ³ : P (41.0) R (39.4) F1 (42.8), V: F1 (41.4) Hom (40.3) Comp (42.5)
				T-REx DS	ARI (20.1), B ³ : P (32.9) R (29.7) F1 (36.8), V: F1 (32.4) Hom (30.1) Comp (35.1)
OHRE (Zhang et al., 2021)	CNN, Virtual Adversarial Training, Reconstruction Loss, Dynamic Hierarchical Triplet Loss, Louvain	word emb, hierarchical information	4.3	FewRel Hierarchy	ARI (64.2), B ³ : P (64.5) R (77.7) F1 (70.5), V: F1 (76.7) Hom (73.8) Comp (79.9)
				NYT-FB Hierarchy	ARI (31.9), B ³ : P (31.4) R (72.3) F1 (43.8), V: F1 (60.0) Hom (49.9) Comp (75.3)
ElementORE (Fangchao et al., 2021)	BERT, T5 (Raffel et al., 2020), Structure Causal Model, K-means	word emb, hierarchical information	4.3	T-REx SPO	ARI (36.6), B ³ : P (46.7) R (43.4) F1 (45.0), V: F1 (45.3) Hom (45.4) Comp (45.2)
				T-REx DS	ARI (25.0), B ³ : P (40.2) R (45.9) F1 (42.9), V: F1 (47.3) Hom (46.9) Comp (47.8)
RoCORE (Zhao et al., 2021)	Relation-oriented Representation, BERT, K-means	word emb	4.3	FewRel	ARI (70.9), B ³ : P (75.2) R (84.6) F1 (79.6), V: F1 (86.0) Hom (83.8) Comp (88.3)
DEEPEx (Wang et al., 2021a)	BERT, Attention, Beam Search, Contrastive Pre-training	NP chunks, word emb, triple emb	4.4	OIE2016 WEB NYT PENN	AUC (58.6), F1 (72.6) AUC (82.4), F1 (91.2) AUC (72.5), F1 (85.5) AUC (81.5), F1 (88.5)

Table 3: Milestone models and representative recent models. ("V" denotes "V-measure", and "emb" stands for "embedding".)